

2. Motivation and requirement (300-400 words, 6 marks)

Tips:

1. Motivation is the reason why the authors propose the solution. It would be the challenge of the domain. Normally, this can be found in the abstract or introduction.
2. Functional/Non-functional requirements are typically summarised by the author. But for the papers without an explicit summary of requirements, you need to extract/derive requirements based on your understanding of the problem/challenge described in the paper, the design of the proposed system, and the domain.
 - a. When documenting requirements try to use keywords like SHOULD, MUST, and MAY. See <https://www.ietf.org/rfc/rfc2119.txt>
 - b. Requirements should be independent of the solution as much as possible.
 - c. Requirements should be documented as complete sentences.
3. You don't have to identify all requirements, just the core ones.
4. Suitability analysis is based on the framework learned in class and tutorial.
5. While each question has its word limit you don't need to write each answer to its fullest length.

2(a) What do the authors aim to solve? 1 marks – 0.25 x 4 points)

The paper is aimed to build a **multi-tenant blockchain system** with a **scalable platform architecture**, which ensures **data integrity** while maintaining **data privacy and performance isolation**.

2(b) What are the main requirements to be addressed while solving the problem?

- Functional requirements (1.5 marks)
- Non-functional requirements (1.5 marks)

Functional requirements (1.5 marks – 0.5 x any 3FRs)

FR1 – The platform should provide the function of writing data. The writing permission is restricted to the platform owner.

FR2 – The platform should provide a cost-efficient writing function that supports writing batches of data.

FR3 – The platform should provide a function to read all the historical transactions/data.

FR4 – The platform should provide functions that allow end users to validate the originality of the data (that is who wrote the data).

FR5 – The platform should provide functions that allow independent agencies to access data for auditing for individual tenants.

FR6 – The platform should support multiple tenants to serve their end users with different business requirements.

Non-functional requirements (1.5 marks – 0.5 x any 3 NFRs)

NFR1 – Integrity: Data integrity of item IDs added by a tenant must be ensured.

NFR2 – Scalability: Scalability at the individual tenant level so that every tenant could store a large volume of data within a period.

NFR3 – Scalability: Scalability at the platform level so that the platform could support several tenants.

NFR3 – Data Privacy: A tenant must not be able to read data of other tenants, like the unique item IDs created, scan event counts, timing, or locations.

NFR4 – Performance Isolation: The tenant with potentially higher workload (e.g., commodity goods with millions of events daily) should not affect read/write performance for other tenants.

2(c) Why is blockchain a good fit for solving this problem? (2 marks – 0.3 x any 5 aspects + 0.5 for decision)

- **Is multi-party required?** Laava's scenarios deal with supply chains with multiple parties. Every tenant is a party.
- **Is operation centralised?** Operation in this context can be loosely coupled between multiple parties. But in the proposed solution the operation is centralized and provided by the platform owner.
- **Is immutability required?** Transaction history and data immutability are desired to enable traceability back to the origin of items and to control fraud and substitution.
- **Is high performance required?** The time taken in Laava's scenarios is dominated by physical transportation and storage. There is no requirement for extreme throughput or latency.
- **Is data transparency required?** Data transparency is required by the parties within a tenant to support logistics planning and identify and respond to problems. Data confidentiality is required between tenants.

Therefore, the given problem is suitable to be solved using a blockchain.

3. The overall design of the solution (~300 words, 5 marks)

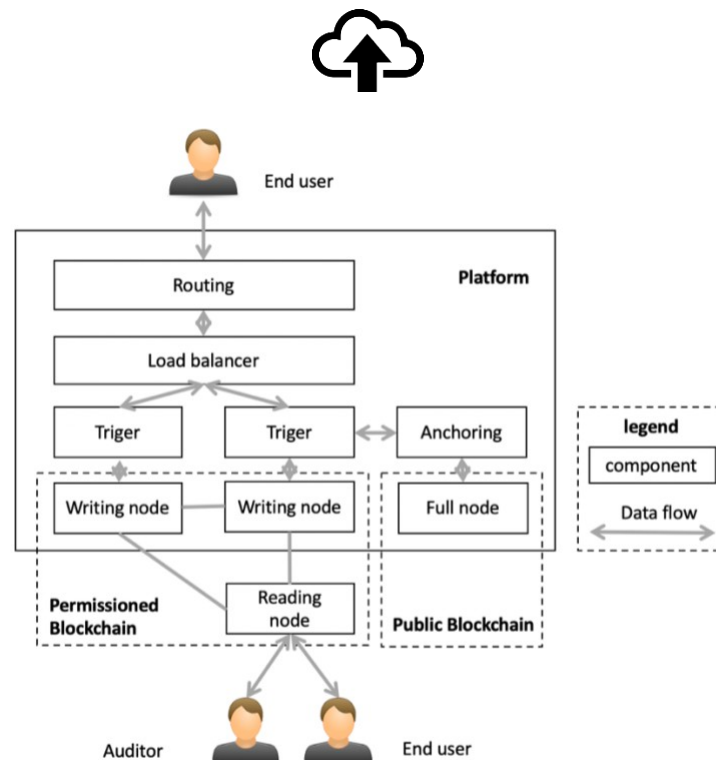
Tips:

1. (Re)draw the main architecture diagram based on your understanding of the system. You could remove some details from the original diagram(s) or add more details if you think the original diagram doesn't cover enough details.
2. A technical architecture diagram with software components is expected (rather than the use case, class diagrams, or sequence diagrams).
3. Consider making the component name bold to make the explanation clearer.

4. Normally, you can find the explanation of all the components within the proposed solution. If the interaction between components is not articulated in the paper, use your judgement, and clearly state any assumptions.
5. Try to use standard notations and make sure the diagram is clearly visible and of high resolution (photo/scan of a diagram drawn on paper is not recommended)

3(a) Draw a diagram of the proposed architecture and upload it here: (3 marks – 0.75 marks for key elements of diagram & connections +0.75 for labelling)

- What are the components?
- How do the components interact?
- A legend is required



3(b) Briefly explain each of the components in the uploaded architecture diagram and comment on them, for e.g., (2 marks – 0.5 x 4 components)

- If an existing blockchain is used? If not, what is the design of the new blockchain?
- What is on-chain, and what is off-chain?
- What off-chain component is used (if any), and for what purpose?

In the proposed architecture, every tenant has an individual permitted blockchain to store their own data. The platform owner hosts the nodes with writing permission to the permitted blockchain for producing blocks while the tenants/auditors host read-only nodes.

An **end user** sends write requests through the **routing component**, which forwards the requests to the respective tenant's blockchain to process. The load balancer distributes the workload to the trigger components associated with different writing nodes. The **end user** and **auditor** send read requests to the read-only node through a public API.

Other than writing data to the permissioned blockchain, the **trigger** is also communicating with the **anchoring** component, which queries the **trigger** to get the Merkle tree root of the tenant's permissioned. The **anchoring component** connects to a node in the public blockchain network, and one node for each tenant's blockchain to be anchored.

A custom **Merkle tree** is designed, where the leaf nodes represent the root of each tenant's permissioned blockchain's state Merkle tree. The custom Merkle trees are stored on each individual blockchain and the root of the custom Merkle tree is placed regularly on a public blockchain through the **anchoring component**.

The **anchoring protocol** starts with querying the latest anchored Merkle root stored on the public blockchain and verifying the Merkle root against the tree maintained in the **anchoring component** to make sure the anchoring component is up to date. For each tenant's permissioned blockchain registered with the anchoring component, there is a subprocess where the protocol queries the blockchain Merkle roots on the latest block. If the chain is in the Merkle tree of roots, the value for the leaf node is updated. If not, the chain is a new tenant, a new node is added to the Merkle tree of roots.

4. Evaluation / performance analysis of the solution design (~250 words, 4 marks)

Tips:

1. Usually, you can find the evaluation section in the second half of the paper.
2. You need to identify the methods used in the evaluation section and the properties evaluated.
3. You need to summarize the setup/process of the experiment, and the result of the experiment. Look for aspects like whether the solution was simulated or had a real implementation, what types of data (real or synthetic) and smart contracts were used, and how the data were collected.
4. Identify the performance metrics and how they were measured.
5. Note your own observations from graphs and tables. Authors may not discuss all findings though they may be in the paper.
6. Use your judgement in interpreting tables/graphs and clearly state any inconsistencies you observe or any assumptions you make.

4(a) How is the proposed solution evaluated? (2 marks – 1 for method + 1 for properties)

- Methods: Experiment, simulation, case study, interview
- Properties: Security, reliability, cost, performance, etc.

The authors developed a research prototype for Laava company, which is a third-party item tracking service provider. Laava provides a unique ID for individual item tracking with various interesting features.

The proposed design is first **examined against the identified requirements** for multi-tenant blockchain-based systems. Second, a **qualitative analysis** is conducted by comparing the proposed architecture with two architecture design alternatives, including one alternative using a global chain anchoring to a public blockchain, and the other alternative using a public blockchain. These three design alternatives are compared against **data integrity, cost, data privacy, performance isolation, availability, and configuration flexibility**.

Third, a quantitative analysis is conducted by measuring the **throughput** of unique ID creation under several conditions and evaluating the time of the anchoring process.

4(b) What were the main findings from the results? (2 marks)

For the throughput evaluation, **4 tests are designed**, including normal load scenario (<15tps), one tenant chain, boundary load scenario (~18tps), one tenant chain, and overload scenario (~18-25tps), one tenant chain, and an overload scenario with three tenant chains (~18-25tps on each chain). The fourth test is aimed to investigate the performance isolation between tenant chains as well as the anchoring protocol.

The results of the 4 tests show that the research prototype can register unique IDs successfully and efficiently, and the **performance is not impacted** by the increased number of tenant chains. For the performance of the anchoring protocol, the total time from the start to the end of each anchoring round is measured. The result shows that the **anchoring times are not affected** by the load of the tenant chain.

5. Potential extensions (~250, 5 marks)

Tips:

1. You might not find the answers to these questions in the paper. If at all discussed, it is either under discussion or summary sections of the paper. The questions in this section focus on your critical analysis of the overall paper based on the material you learned in class.
2. Whether an answer is right or wrong depends on your arguments and assumptions. So clearly state any assumptions.
3. In the case you find the “strengths” claimed by the authors in the paper, it’s still ok to argue them as weaknesses. For example, you can argue that using individual permissioned blockchain for each tenant increases the complexity of the platform, resource utilisation, and maintenance cost.

4. Based on the arguments you build for strength/weakness, suggested improvements can be different and again depend on arguments.
5. Limit your discussion to technical aspects relevant to the paper, e.g., **do not** comment on the writing style of paper or clarity of figures. Also, for a problem that does not need high throughput, it does not make sense to suggest that authors improve throughput. Also, do not focus on aspects that are not related to the class, e.g., problems with the use case.
6. No marks will be given to generic statements like increase throughput, scalability, or reduce latency without a clear reason to do so and some points on how it can be done.

5(a) Provide a critique of the paper: (3 marks – 1 for strengths + 1 for weaknesses+ 1 for evaluation)

- What are the technical strengths of the proposed solution?
- What are the technical weaknesses of the proposed solution?
- Will the evaluation convince potential users to adopt the proposed solution?

Strength

- Every tenant in the proposed solution has a dedicated permissioned blockchain to maintain their own data and smart contract. Such a design **provides data privacy** for the tenant.
- Using multiple blockchain rather than a global blockchain **improves the overall performance and scalability**.
- Using permissioned blockchain rather than public blockchain could minimize cost.
- Using a customized Merkle tree and anchoring with a public blockchain can **achieve data integrity for data with arbitrary** size.

Weakness

- Writing operations to all the tenant chains is through the Laava platform, which **becomes a single point of failure** from a software architecture perspective. Although the routing component and load balancers can mitigate the workload bottleneck from a technical perspective.
- The anchoring component is a centralized component, which is maintained by the Laava platform. The whole anchoring process also happens on the platform. Such central operation becomes a potential single point of failure.

5(b) How would you improve the proposed solution? For e.g., (2 marks – 1 x 2 points)

- Replace a component to improve a property.
- Apply a design pattern to achieve a property.

- One potential improvement is to further decentralize the architecture by allowing the tenants to maintain the writing operations by themselves.

To do this, tenants need to host their admin nodes with permission to write data.

- Another potential improvement is to enable self-regulation through implementing governance and regulation rules through the smart contract.
- In the case of using a more decentralized architecture, multiple [authorization pattern](#) could be applied when the cryptographic representation on public blockchain needs to be updated.